

Linear Algebra Final Project

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Abstract

This project explores the Least-Squares Monte-Carlo Simulation Approach (LSM) for pricing American Options. We focus on its foundations of linear algebra and numerical computing to replicate the results in the paper, said Valuing American Options by Simulation: A simple Least-Squares Approach (Francis A. Longstaff; Eduardo S. Schwartz, 2001), with the given data. We model the valuation problem as a projection of future cash flows on a finite subspace spanned by basis functions. Besides testing the correctness of our algorithm, we investigate the numerical stability as well. In the empirical studies, we demonstrate the trade-off between the theoretical accuracy in higher-degree polynomials and the numerical instability as well as inaccuracy resulting from the increase of condition number of design matrix.

1. Introduction

In the course of derivatives pricing, we only answer how to value European options. Since it can only exercise in maturity, we do not need to care about what are the paths before maturity. However, American Options may be exercised whenever we want. Hence, the notable Black-Scholes-Merton Model cannot be used for it. To determine the optimal stopping time, we found the LSM algorithm proposed by Longstaff and Schwartz in 2001 which may deal with the optimal stopping problem by using cross-sectional data to estimate the conditional expected value of continuation.

From linear algebra perspectives, this method simplifies the issue from a complicated stochastic problem to a classic and understandable linear least-squares problem.

$$Ax \approx b$$

Applying the concepts in orthogonality which were covered in linear algebra course this semester, we solve this approximation problem efficiently and compute the solution with Python.

The objective of this project is to implement LSM algorithm and, more importantly, analyze the impact of basis function selection (polynomial degree) on the numerical stability of the matrix operations. We aim to demonstrate how linear algebra concepts, such as condition number, play a vital role in the accuracy and robustness in financial modeling.

2. Methodology

2.1 Modeling the Underlying Asset

We model the stock price dynamics under the risk-neutral measure using Geometric Brownian Motion (GBM). In our numerical implementation, we discretize the time interval $[0, T]$ into M steps of size Δt . For I independent simulation paths, the stock price evolution is computed using vectorized matrix operations:

$$S_{t+1} = S_t \odot \exp \left[\left(r - \frac{1}{2} \sigma^2 \right) \Delta t + \sigma \sqrt{\Delta t} Z_{t+1} \right]$$

where:

$S_t \in R^I$: The vector of stock prices at time t .

$Z_{t+1} \in R^I$: Standard normal random vector.

$\odot$ denotes element-wise multiplication (Hadamard product).

2.2 Linear Algebra: Projection onto Finite-Subspace

The core of the American option pricing problem is estimating the continuation value

$$E[e^{-r\Delta t} V_{t+1} \mid S_t].$$

From a linear algebra perspective, we treat this conditional expectation as an unknown function $F(S_t)$ residing in an infinite-dimensional function space. We approximate $F(S_t)$ by projecting it onto a finite-dimensional subspace $\mathcal{V}$ spanned by a set of basis functions $\{ \phi_0, \phi_1, \dots, \phi_K \}$.

For the paths that are in-the-money at time t , we model the relationship as a linear system:

$$Y = A\beta + \epsilon$$

where:

Target Vector $Y \in R^{N_{itm}}$: The discounted cash flows from time $t+1$ for the N_{itm} in-the-money paths.

Coefficient Vector $\beta \in R^{K+1}$: The unknown weights of the basis functions.

Error Vector ϵ : Represents the simulation noise and approximation error.

2.3 Construction of the Design Matrix

We construct the Design Matrix $A \in R^{N_{itm} \times (K+1)}$ using polynomial basis functions. If we choose a polynomial basis of degree K , A takes the form of a Vandermonde Matrix:

$$A = \begin{matrix} & 1 & S_1 & \cdots & S_1^K \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ & 1 & S_N & \cdots & S_N^K \end{matrix}$$

where S_i represents the stock price of the i -th path at time t . Each column of A represents a basis vector in $R^{N_{itm}}$.

2.4 Least Squares Solution via Normal Equation

To find the optimal approximation, we minimize the squared 2nd-norm (Euclidean) of the residual vector $\|\epsilon\|^2 = \|Y - A\beta\|^2$. This is a classic Linear Least Squares problem. Using vector calculus, the analytical solution is given by the Normal Equation:

$$A^T A \tilde{\beta} = A^T Y$$

Assuming A has full column rank, the optimal coefficient vector is:

$$\tilde{\beta} = (A^T A)^{-1} A^T Y$$

Geometrically, the estimated continuation value $\tilde{Y} = A\tilde{\beta}$ represents the orthogonal projection of the realized cash flow vector Y onto the $C(A)$.

2.5 Numerical Stability Analysis: Condition Number

An important goal of this project is analyzing the robustness of the linear system. As the degree of the polynomial basis K increases, the columns of the Vandermonde matrix A tend to become nearly linearly dependent. We measure this instability by Condition Number $\kappa(A)$:

$$\kappa(A) = \|A\| \cdot \|A^{-1}\| = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}$$

where $\sigma_{\max}(A)$ and $\sigma_{\min}(A)$ are the largest and smallest singular values of A . A large condition number implies that the matrix $A^T A$ is ill-conditioned which means the solution is very sensitive to perturbations, i.e. it's very unstable. In such cases, small perturbations in the simulation data Y can lead to terrible errors in $\tilde{\beta}$, causing the pricing algorithm to diverge or produce unreliable results.

3. Empirical Studies

3.1 Replication of the paper

To validate the correctness of our vector-based implementation, we first replicated the numerical example provided in the original paper. The example involves pricing an American Put Option using 8 simulation paths over 3 time steps. Take $t=2$ for example:

```
--- Time Step t=2 ---
ITM Paths Indices (0-based): [0 2 3 5 6]

[Matrices Visualization]
Matrix A (Basis: 1, X, X^2):
[[1.          1.08         1.1664001 ]
 [1.          1.07         1.1449001 ]
 [1.          0.97         0.9409      ]
 [1.          0.77         0.5929      ]
 [1.          0.84         0.70559996]]
Target Vector Y (Discounted Cashflow):
[0.          0.06592357 0.16951764 0.18835296 0.08475885]
Coefficients:
[-1.0699897  2.983416  -1.8135796]
-----
Estimated Continuation Value:
[0.03674054 0.0458982 0.11752701 0.15196943 0.15641809]
Immediate Exercise Value:
[0.01999998 0.02999997 0.13          0.33000004 0.26000005]
Decision (True = Exercise):
[False False True True True]
```

Figure 1: Numerical Results of the Replication

As shown in Figure 1, our algorithm produced the regression coefficients $[-1.0699897, 2.983416, -1.8135796]$ which is identical to coefficients of conditional expectation function at the middle of p117 in paper. Furthermore, the exercise decisions derived from our comparison of

immediate exercise value versus continuation value are identical to the published results. This confirms that our construction of the Design Matrix A and the least-squares solver are correctly implemented.

3.2 Stability Analysis: Trade-off Between Accuracy and Stability

We further conducted a large-scale simulation ($I=50,000$ paths, $M=50$ steps) to analyze the impact of the polynomial degree K on the robustness of the linear system. The results are visualized below:

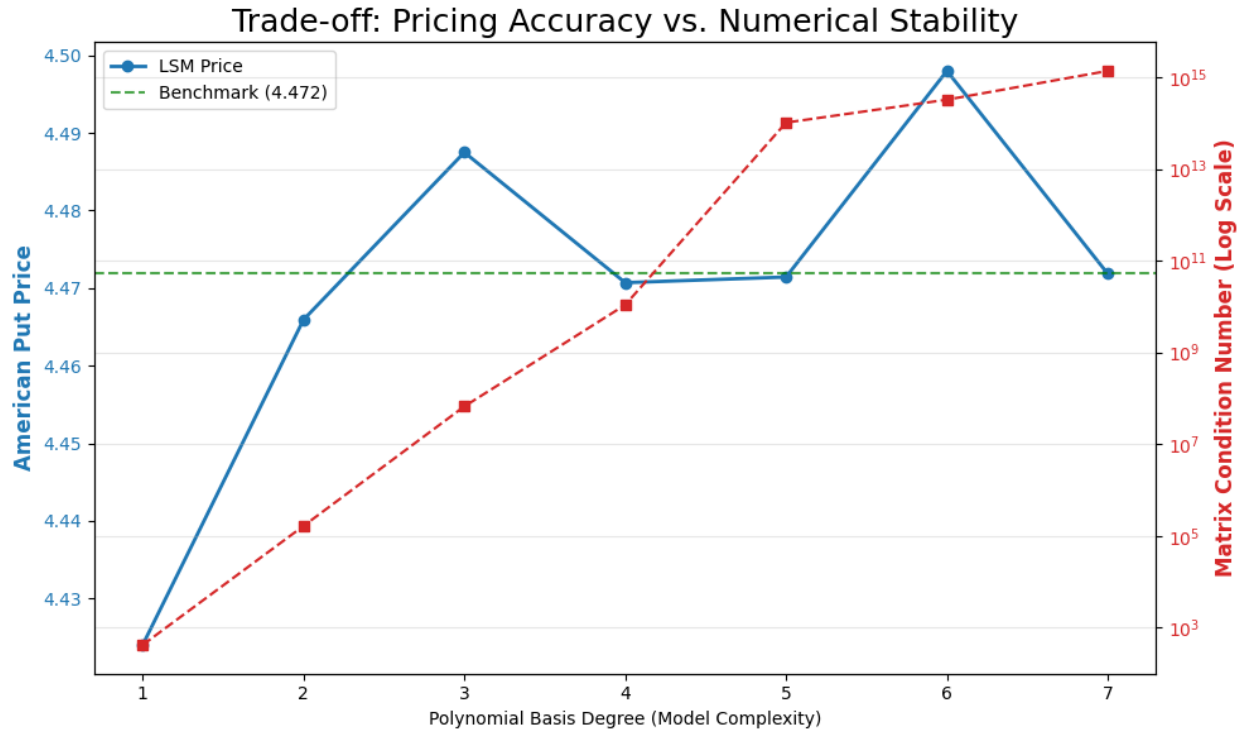


Figure 2: The Trade-off between Pricing Accuracy and Numerical Stability

The results in Figure 2 reveal a critical insight from a linear algebra perspective:

1. **Pricing Accuracy (Blue Line):** The option price converges to the benchmark value 4.472 (provided in the paper) as we increase the basis degree from 1 to 3. This indicates that a linear basis is insufficient to capture the curvature of the value function (underfitting), while a cubic basis provides a satisfactory approximation.
2. **Matrix Instability (Red Line):** However, the condition number $\kappa(A)$ increases exponentially with the degree K . At $K=3$, condition number is approximately 10^5 which is numerically undercontrol. At $K=7$, condition number is approximately 10^{13} which is extremely unstable.

3. The "Divergence" Point: Crucially, when $K=7$, the pricing accuracy begins to degrade, the blue line drops away from the benchmark. This confirms that the Design Matrix A has become ill-conditioned. The basis functions $\{1, S, \dots, S^7\}$ exhibit severe multicollinearity, causing the matrix $A^T A$ to become nearly singular. Consequently, numerical noise in the simulation is amplified, overriding the theoretical benefits of a higher-degree approximation.

3.3 Short Conclusion of Empirical Studies

Our empirical results suggest that a cubic polynomial ($K=3$) offers the optimal balance between reducing approximation bias and maintaining numerical stability for this specific pricing problem.

4. Conclusions and Future Works

4.1 Conclusion

In this project, we successfully implemented the LSM algorithm for American option pricing and verified its correctness by replicating the results of Longstaff and Schwartz (2001). From a linear algebra perspective, we demonstrated that the valuation problem is fundamentally a projection of future cash flows onto a finite subspace.

Our empirical analysis highlights a crucial trade-off between model complexity and numerical stability. While higher-degree polynomial bases theoretically reduce approximation bias, they lead to an exponentially increasing Condition Number of the design matrix. This ill-conditioning amplifies simulation noise, rendering high-degree approximations unreliable. We conclude that a cubic basis ($K=3$) offers the optimal balance for this specific problem.

4.2 Future Works

To address the issue of ill-conditioned matrices, future research could replace the standard monomial basis $\{1, x, x^2, \dots\}$ with Orthogonal Polynomials, such as Laguerre or Hermite polynomials. Since orthogonal basis vectors satisfy $\phi_i \cdot \phi_j = 0$ for $i \neq j$, the resulting Gram matrix $A^T A$ becomes diagonal (or nearly diagonal). This would significantly minimize the condition number $\kappa(A) \approx 1$, allowing for higher-order approximations without sacrificing numerical stability.

Reference:

Longstaff, F. A., & Schwartz, E. S. (2001). Valuing American Options by Simulation: A Simple Least-Squares Approach. *The Review of Financial Studies*, Vol. 14(1), p113-147.

Appendix:

Python Code on Google Colab:

<https://colab.research.google.com/drive/1p6Y6kyy46SXLiY25GBqhONVcqIHSeIWO?usp=sharing>